

Spatio-Temporal Demographic Analysis of US Nuclear Fuel Cycle Facilities

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INTRODUCTION

Throughout the development of the modern-day nuclear power industry, countless promises have been made to the industry's ability to spur vast social and economic progress. Ready access to energy is generally considered to be a cornerstone of societal development, and the energetic capabilities of the atom have only amplified assurances of availability.

Despite this, there have been innumerable injustices caused by the US nuclear fuel cycle (NFC), the lifeblood of nuclear power. Mines have been left uncovered and unremediated without proper decommissioning¹ radioisotopic releases have been covered up² and waste siting efforts have frequently occurred in and around American Indian tribes^{3,4}. This begs the question, *"To what extent has the US NFC impacted different groups of people?"* Thus, the primary objective of this work is to quantify any potential injustices caused by the US NFC.

This research produces two main novel findings. The first is a consolidated database of all commercial US NFC facilities and their coordinate locations, including milling and ore processing, conversion, enrichment, fuel fabrication, power operation, and on-site waste storage, including the emergence of these facilities over a 70 year period. This database also includes the locations of considered waste locations, for both permanent and interim storage purposes, in addition to undated mines.

The second is a comparative analysis of risk and harm imposed on different socioeconomic demographics by the US NFC. This comparative analysis is in an effort to quantify the historic sentiments behind claims of injustice that have been applied to the nuclear industry (environmental racism[†], classism[‡], among others). This analysis takes census data near NFC facilities at their time of siting, in addition to present-day demographics, to nominally estimate risk over time and space across different census-designated demographics.

[†] *"The slow pace of [uranium mine] cleanup is directly related to the law, itself ... The law places more importance on the relationship between EPA and the companies that caused the problem than it creates a right of sitting at the table of the local affected community. And so on Navajo that is institutional racism."*⁵

The results of this research have a wide range of applicability, namely for individuals concerned with NFC operations, policymakers who work along the NFC, particularly those looking to site new facilities such as interim spent fuel storage facilities, deep geologic repositories, and new reactors. The location database could be used to inform communities across the NFC as to what exact facilities are near them as well as their associated risks. This database may also prove useful for public oversight purposes, as any individual (planner, policymaker or citizen) would not have previously been able to view the entire US nuclear inventory. The demographic analysis may be used to support a wholesale cost-benefit analysis of the NFC (i.e., the benefit of the consumers of nuclear-generated electricity compared to the risks/harms/costs imposed on those along the NFC). Regardless of any future application, this investigation provides a necessary insight into the interactions between the US NFC and its surrounding social contexts.

METHODS

There are three primary methodological aspects to this work. The first concerns the identification and location of NFC facilities in the US. The second pertains to the characterization of risk of each individual facility around the NFC. The third relates demographic data, over time and space, to the associated risk of NFC facilities. We performed all mapping and demographic data pre-processing in the Geographic Information System ArcGIS by ESRI, and retrieved all demographic data and county geography from the National Historical Geographic Information System⁷.

Location Data

NFC facilities were separated into three categories. These categories consist of front-end (mining, processing, conversion, enrichment, fabrication), power operation, and (high level radioactive) waste storage. With the exception of

[‡] *"Minority and low-income communities are at higher risk of the radiation pumped via nuclear waste into their communities because of their proximity. At the same, these communities have statistically higher levels of women and children. These risk factors, much like the reasons nuclear power plants are built in these areas in the first place, perpetuate racist and classist outcomes."*⁶

mines, the names and operational dates (up to roughly 2012) of front-end facilities were available from the International Atomic Energy Agency's (IAEA) Nuclear Fuel Cycle Facility Database (NFCDB)⁸. With these names and dates, 102 geographic locations were corroborated manually from 80 unique sources, including environmental impact statements, public records, or, on occasion, simply as they were noted on existing Global Positioning Systems. Operational dates were updated where applicable. The exact locations of uranium mines were available from a 2006 US Environmental Protection Agency (EPA) database⁹, which itself was compiled from 24 other databases. While extensive, it is necessary to note that these databases are entirely complete due to the large number of small, undocumented mines. Thus, any data derived from mining facilities should be considered as a conservative estimate.

A complete list of names and operational dates of nuclear power reactors were likewise extracted from an IAEA database, the Power Reactor Information System (PRIS)¹⁰. A total of 134 reactors (operational and shut down) were located at 85 unique sites through the same means as the front-end facilities. Research reactors are not included in this count.

Waste storage installations were located in a different manner from the previous facilities, as this investigation considers the *proposed sites*, not simply where waste currently lies. The NFCDB lists 56 reactor sites with co-located away-from-reactor (AFR) waste installations. Another 38 proposed sites were identified, with 27 interim waste sites and 11 permanent. *Proposed* in this context includes sites at which resources were used to investigate a *specific* site (e.g., resources were spent on studying the suitability of crystalline rock structures for waste disposal, but were not included as these were studies of broad regions as opposed to sites¹¹). Interim site names were taken from ¹², ¹³, and permanent from ¹⁴. Geographic coordinates were located for sites where environmental assessments were performed (namely in ¹³ and ¹⁵). For sites where environmental assessments were not performed, the location was noted as the host county or tribe's census-designated internal coordinates¹⁶.

Risk Characterization

Injustice is defined within this work as the unequal treatment of an individual or group of individuals. Therefore, when quantifying injustice, it is necessary to 1) evaluate the individual's/groups's treatment 2) in relation to another's. This work separates the treatment of individuals by the NFC into three categories: realized harm, nominal risk of hazardous facilities, and the attempted siting of hazardous facilities. At this current stage of research, only nominal risk and attempted siting are considered, as a complete list of realized harms is both to-date incomplete and overall impossible to assess with accuracy or certainty.

At this stage of analysis, risk is taken solely as an upper bound with zero uncertainty. As this work draws conclusions on relative risk and risk sharing, as opposed to absolute risk, upper-level estimates and comparisons among different groups will provide more complete information for investigation. The upper bound limit is taken as the upper limit that is allowed by US regulation, specifically 1 mSv/year from all facilities across the fuel cycle¹⁷.

Demographic Analysis

For initial analysis, demographic data were taken from the 5-year, 2016-2020 American Community Survey. Demographics of interest included race, level of education, and ratio of income to poverty level. Population centers for each county in the US were taken from the 2020 Decennial Census and mapped along with the corresponding demographic data.

The previously identified NFC facilities, excluding mines, were mapped along the demographic county data. This analysis is independent of time and can be considered a cumulative count of all decommissioned, active, and proposed (waste) facilities near the 2020 US population. 50 mile buffer zones (corresponding to Ingestion Exposure Pathway Emergency Planning Zones) were drawn around each facility, and counties whose population centers lie within each buffer contributed their demographic data to the facility's tabulation. Following tabulation across all counties and NFC facilities, we conducted analysis to determine the demographic makeup surrounding each category of facility.

For this analysis, two assumptions were made. The first was that a county's entire population exists at a single point, the population center. This analysis used county-level data, as finer-scale data like census tracts are less available for longer time periods (e.g., 1950-2020). Thus, not having finer data (nor population distributions) necessitates population-averaged point estimates.

The second assumption was that any events that constitute risk surrounding the facilities were independent. The corollary assumption is that the buffer zones surrounding each facility are allowed to overlap freely. This essentially leads to 'double counting' of individuals, and subsequently the risk that they experience. This is a fair assumption, as it is sensible for an individual who lives in multiple buffer zones to experience more risk than the individual who lives in only one.

Demographic information near uranium mines was excluded from this initial analysis. Further review of location data and operational dates is required to limit uncertainty and to generate accurate demographic profiles. It is necessary to note, however, that because of their relatively small geographic footprint and large number of sites (roughly 5,000 mine sites compared to 300 non-mine sites), mines must be treated differently from other NFC facilities. Mines will be counted within each county's borders, as opposed to counties

within a mine's buffer zone, and demographic data will be compiled with the remaining facilities.

RESULTS

The results of the location collection have been mapped using ArcGIS. A collection of figures illustrating this map can be found in **APPENDIX A**.

The results of the demographic analysis can be seen in Tables 1, 2, and 3. Each table lists the demographic composition surrounding front-end facilities (excluding mines), nuclear power plants (NPPs), and back-end facilities (actual and proposed waste storage and reprocessing facilities). Each table also shows the deviation of the demographic to the 2016-2020 national average.

Table 1 describes racial and ethnic distribution surrounding the different facility categories ("AIAN" denotes American Indian or Alaska Native population). Table 2 shows the education level surrounding each category of facility. Table 3 shows the level of poverty (for whom the ratio of income to the national poverty level is less than one).

Table 1. Deviation of racial and ethnic composition across the NFC, compared to national average.

	White Only	Black Only	AIAN only	Asian Only	Hispanic
Front End (excl. mines)	-3%	+11%	-3%	+17%	+6%
NPPs	-1%	-1%	-37%	+8%	+25%
Back End	-2%	+11%	-10%	+11%	+5%

These preliminary results illustrate a number of initial findings. Pertaining to racial and ethnic composition, there is a notably greater concentration of individuals who identify as Hispanic and Asian across each NFC category, and of individuals who identify as Black along the front- and back-end. There is a decreased concentration of individuals who identify as White or AIAN across all facilities of categories. This can generally be interpreted as all non-White (aside from American Indian/Alaska Native) and Hispanic populations experience greater risk from the NFC than their White or non-Hispanic counterparts.

Table 2. Deviation of educational attainment (of ages 25+) across the NFC, compared to national average.

	No school	High School/GED	Bachelor's & up
Front End (excl. mines)	+4%	+1%	0%
NPPs	+14%	-2%	+1%
Back End	+3%	+2%	0%

Table 3. Population under poverty.

	Poverty rate	% Deviation
Front End (excl. mines)	12.21%	-4.95%
NPPs	13.26%	+3.29%
Back End	13.15%	+2.41%

The discrepancies between the analyzed AIAN data and the qualitative context supplied earlier ^(1,2) can be reconciled with three possible explanations. The first is that AIAN populations are highly concentrated in a smaller number of geographic locations, which has the greatest implications for NPPs (additionally, there are no commercial reactors in Alaska). Other races, which are typically more spatially homogeneous than AIAN populations, are located near the likewise sparse and concentrated NPPs. The second is the exclusion of mines from this phase of analysis, which impacts the front-end category. The third is the absence of weighting among proposed waste sites, as efforts for waste siting generally shifted towards tribal land after downselections (e.g. Yucca Mountain for repositories¹⁵, Skull Valley and Mescalero Apache for interim¹⁸).

It can also be seen that there are higher proportions of individuals with no education among each category of the NFC. This generally correlates to poverty rates near NPPs and back-end facilities, but not to front-end facilities. One possible explanation for this is that front-end facilities like conversion, enrichment, and fuel fabrication are reliant on

industry and infrastructure, which may coincide with communities with lower poverty rates. NPPs and back-end facilities tend to exist in more rural zones, contributing to a higher poverty rate.

It is important to note, however, that these are initial results for a single time period and do not reflect uncertainty. The goal of future work is primarily to analyze trends over time and propagate associated uncertainties.

FUTURE WORK

Later work will include demographic analysis surrounding uranium mines in addition to trends over time (from 1950 to 2020). Risk will also be weighted in a manner which is proportional to each facility's capacity (e.g., a 1 GW NPP poses less risk than a 300 MW NPP), and to its seriousness in consideration for proposed facilities (e.g., Yucca Mountain posed a greater risk to its nearby populace than Vacherie Dome). Additionally, the demographics surrounding transportation routes will be explored (i.e., Class 7 Highways), and potentially for research facilities (national labs, universities, military testing sites).

APPENDIX A: NFC LOCATION MAP

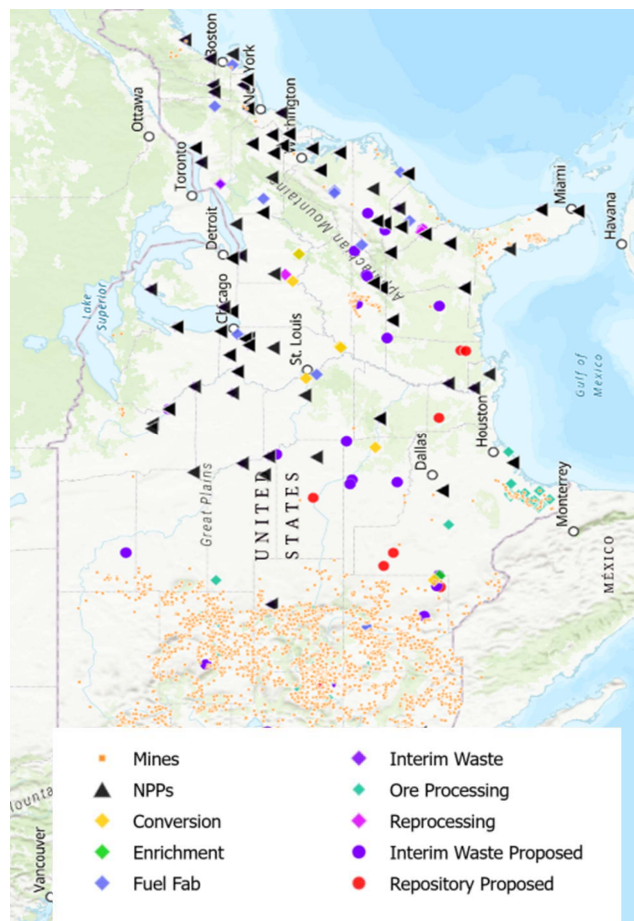


Fig. 1: US NFC Facility Map

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